

João Vitor Dias Fernandes

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Technical Skills

- **Languages:** TypeScript, Python, Java, PHP, JavaScript, Dart
- **Web Development:** HTML, CSS, JSF, REST APIs
- **Databases:** PostgreSQL, Supabase, MySQL, Oracle Database
- **Frameworks:** Angular, FastAPI, Laravel, Spring Boot, Flutter
- **DevOps:** Docker, Gradle, Grafana, Maven
- **Computer Vision:** OpenCV, YOLOv8, MediaPipe, OCR
- **Version Control:** Git, GitHub, GitLab, GitKraken

Professional Summary

Computer Science graduate from the Federal Institute of Minas Gerais (IFMG), with hands-on experience in software engineering and web application development. Experienced in building production systems, real-time industrial monitoring platforms, and solutions involving APIs, automated data processing, and computer vision.

Professional Experience

SensorEng Sep 2025 - Sep 2026
Frontend Developer

- Led the architectural refactoring of a legacy monolith into a modular design, introducing lazy loading and reducing the initial application bundle size by 85%.
- Developed and enhanced real-time industrial monitoring interfaces, integrating continuous video streaming, AI inference results, IoT sensor telemetry, and georeferenced maps.
- Built an internal reusable component library (dynamic data tables, filters, and modals) adopted system-wide, standardizing the user interface and reducing new screen development time by 60%.
- Implemented role-based access control (RBAC) with route guards and authentication, ensuring secure multi-tenant data isolation across client organizations.
- **Context:** Remote position within an agile team delivering an industrial automation and computer vision platform, directly impacting 50+ enterprise clients.

Avante Sistemas Aug 2024 - Jan 2025
Junior Full-Stack Developer

- Developed and maintained features for Enterprise Resource Planning (ERP) software, supporting the evolution of business management modules and workflows.
- Diagnosed and resolved code and database defects, optimizing SQL queries and applying fixes to improve system stability and reliability.
- Contributed to implementing new features and tailoring software capabilities to the operational needs of various company departments.
- **Context:** On-site position at a software company in Formiga, MG, building enterprise management solutions across diverse business segments.

Credit Management Software Mar 2024 - Mar 2025
Software Engineering Intern

- Developed automated document parsing and data extraction services using Computer Vision and OCR (Optical Character Recognition), enabling automatic identification and classification of over 20 document types (including articles of association, financial statements, driver's licenses, and vehicle registrations).
- Contributed to integrating application components and maintaining existing services.
- **Context:** Hybrid role as a scholarship research intern in an extension project at the IFMG Innovation Hub in partnership with the private sector, developing a web platform for commercial credit automation and approval.

IFMG Time & Attendance Mobile App May 2023 - Oct 2023
Software Engineering Intern

- Developed a cross-platform mobile application integrated with SUAP (Unified Public Administration System), implementing script injection for automated authentication and time-clock registration.
- Built real-time web scraping routines to retrieve and display daily attendance history (clock-in/clock-out events), reducing discrepancies in attendance records.
- Implemented secure local credential storage, a notification scheduling module with audio reminders, and institutional Wi-Fi validation to ensure valid attendance registration.
- **Context:** Work at the IFMG Innovation Hub, developing a mobile application for daily work hours and attendance tracking for campus faculty and staff.

Projects

Russian Mastery

Full-Stack Project

- Architected and developed a full-stack Russian language learning platform, implementing the SuperMemo-2 (SM-2) spaced repetition algorithm for dynamic review scheduling and long-term memory retention.
- Built an interactive frontend featuring an on-screen Cyrillic keyboard simulator, interactive practice exercises, and a gamification system with XP and level progression.
- Engineered a RESTful API leveraging Redis for query caching and PostgreSQL for storing a database of over 43,000 words and sentences.
- Implemented secure authentication with JWT (JSON Web Tokens), role-based access control (RBAC), and containerized the entire ecosystem with Docker.
- **Context:** Personal project combining Russian language learning with full-stack development, featuring phonetic stress-mark processing, audio playback, and multi-theme support.

Computer Vision Pipeline for Boxing Matches (Bachelor's Thesis)

Researcher & Developer

- Designed and implemented a computer vision pipeline for real-time human pose estimation and athlete tracking, training and fine-tuning YOLO (You Only Look Once) models on a custom annotated dataset.
- Developed geometric segmentation modules with dynamic ROIs (Regions of Interest) for athlete body tracking, transitioning from clustering to direct custom model detection and reducing total processing time by 83.6%.
- Modeled a finite-state machine (FSM) to automate the detection and counting of valid punches and fouls.
- Integrated a companion mobile application with an asynchronous backend server, enabling live video streaming directly from a smartphone camera.
- **Context:** Bachelor's thesis in Computer Science at IFMG, combining artificial intelligence, real-time video processing, and sports analytics.

Education

Federal Institute of Minas Gerais (IFMG)

Computer Science

Bachelor's Degree

Jan 2020 - Aug 2025

Languages

Portuguese:	Native
English:	C1 (Advanced)
Spanish:	B1 (Intermediate)
Russian:	A1 (Beginner)